

Aaron Xiong

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EDUCATION

University of Michigan

Ann Arbor, MI, USA

Bachelor of Science in Engineering in Computer Science, Minor in Electrical Engineering

May. 2027

Cumulative GPA: 3.94/4 (Dean's Honors List)

(Expected)

Courses: Embedded Systems, Computer Security, Data Structures and Algorithms, Computer Organization, Linear Algebra.

SKILLS

- **Programming Languages:** C/C++, Python, Swift, HTML/CSS, JavaScript, MATLAB, Verilog, ARMv7, NIOS II, Arduino, SQL
- **Libraries & Frameworks:** ReactJS, Flask, Tailwind CSS, Bootstrap, PyTorch, NumPy, PySerial, Matplotlib, GTK, Libcurl
- **Tools:** Linux Ubuntu, Git, Docker, Oracle, VMWare, GDB, Wireshark, Ghidra

WORK EXPERIENCE

NLP Research Assistant at LAUNCH Lab

Mar. 2025 – Present

- Developed a comprehensive Open-source LLM benchmark on Olympiad-Level Code Generation with **403** Informatics Olympiad problems from **72** competitions between 2023 and 2025, featuring hidden test suites and human contestant rankings.
- Extracted, restructured, and processed task files for each competition and built a contestant database by locating profiles with Serper and parsing Codeforces rating histories, using Google Gemini AI for profile matching based on name, country, and text.
- Evaluated performances of **20+** LLMs against human contestants and systematically computed Elo ratings for each LLM model to benchmark code generation capabilities based on test suite results.

Full-stack Developer for U-M School of Nursing

Jan. 2025 – Dec. 2025

- Designed a HIPAA-compliant, mobile-friendly, Full-stack web platform with a team of seven that aims to provide Nurse Health Coaching, Educational Content, and Personalized Community Resources for pregnant people and infant caregivers.
- Integrated Zoom and Qualtrics for nurse coaching and patient assessments, Google Maps for geospatial analysis, Oracle Database for patient data storage, and U-M's text-based generative AI tool Maizey for summarization functionalities.
- Engineered and optimized prompts for Maizey to accurately summarize meeting notes from nurse-patient Zoom sessions, generate personalized care plans for clients, and extract key needs from health surveys while mitigating hallucinations to ensure safety.
- Designed an accessible and intuitive UI with Figma through interviews and usability testing, ensuring a seamless user experience.

PROJECT EXPERIENCE

Armed (Embedded System)

Oct. 2025 – Dec. 2025

- Designed an auto-aim embedded system for a projectile launcher using an STM32 Nucleo-L4R5ZI-P MCU and an IR Camera.
- Leveraged interrupt-driven firmware architecture to efficiently achieve real-time sensor input processing and system state updates.
- Implemented firmware and circuitry for haptic motors, LED boards, and piezo buzzers, providing multimodal aiming guidance.
- Provided a rich interface with buttons and an LCD character display, and configured Xbee modules for wireless haptic feedback.
- Modeled and 3D-printed a custom mounting platform, securing all electrical components onto the projectile launcher.

LockeDown (Computer Vision, LLM, Python)

Sep. 2025

- Developed a Full-stack real-time posture and productivity monitoring system integrating React, FastAPI, and SQLite.
- Leveraged YOLOv8m-pose and OpenCV to build a Linux program that detects slouching and delivers audio alerts to the user, integrating MemryX accelerators to optimize video inference and adopting a multi-threaded architecture for seamless performance.
- Monitored application sessions and classified productivity with Gemini, caching past sessions with SQLite3 to reduce token usage.
- Created a gamified scoring system with leaderboards, historical analytics, and real-time dashboards to encourage user engagement.

Evie (C++, GIS, Algorithms)

Jan. 2024 – Apr. 2024

- Developed a Geographic Information System in C++ for electric vehicle users across multiple regions with a team of three.
- Extracted real-world map data from StreetsDatabase and OpenStreetMapDatabase, rendered visuals using EZGL and GTK, integrated live weather data with Libcurl, designed a user-friendly interface using Glade, and managed team data with Git.
- Implemented A* Search for efficient navigation functionalities in the application and a TSP algorithm to optimize delivery routes, reducing the average path length by about **20%** relative to the initial implementation and ranking **top 15 out of 91** submissions.
- Led collaborative coding efforts by strategically dividing and distributing tasks and tracking team progress with Agile Roadmaps.

PUBLICATIONS

- Zou, K., Xiong, A., Zhang, Y., Zhang, F., Ren, Y., Yang, J., Lee, A., Bhushan, S., & Wang, L. (2025). *LiveOIBench: Can Large Language Models Outperform Human Contestants in Informatics Olympiads?* Preprint, <https://arxiv.org/abs/2510.09595>